

# YOLO-C3AC: An Efficient Deep Learning Method for CT Image Detection of Kidney Stones

Jingqing Tang\*  
School of Computer Science and  
Technology  
Chongqing University of Posts and  
Telecommunications  
Chongqing, China  
1220573267@qq.com

Yonghong Li  
School of Science  
Chongqing University of Posts and  
Telecommunications  
Chongqing, China  
liyh@cqupt.edu.cn

Juan Wang  
Department of Ultrasonography  
The Fifth People's Hospital of  
Chongqing  
Chongqing, China  
42283936@qq.com

## Abstract

Nephrolithiasis is a structure formed by a solid concentration or accumulation of crystals in the kidneys. With lifestyle changes, the prevalence of kidney stone disease is gradually increasing. Computed tomography (CT) is a commonly used imaging tool for kidney stones, CT images of kidney stones have low resolution, overlapping organs and tissues, small target stones and distribution, aggregation or discreteness, manual detection and analysis of kidney stone images will be affected by subjective judgment and personal experience, especially the analysis of inexperienced doctors may lead to a high error rate, and the efficiency is relatively low in actual detection. Machine learning based methods focus more on whether there are stones in the image, rather than the location information of the stones. Although general deep learning methods can detect the approximate location of stones, it cannot fully detect the specific number and size of the stones in the image. In the complex background of CT images of kidney stones, the YOLO series algorithm can achieve better detection results. Our YOLO-C3AC has improved the original YOLOv5, with the C3AC module being able to obtain adaptive contextual information, thereby achieving improved model detection accuracy. FusionBlock can improve the model's ability to perceive multi-scale detection of target spatial positions, thereby enhancing the model's ability to detect target stones. The MPDIoU loss function can improve the speed of boundary regression, thereby enhancing convergence speed and improving detection accuracy. Soft NMS can reduce the probability of false positives and missed detections, improve accuracy and efficiency. Compared with YOLOv3, YOLOv5, YOLOv7, YOLOv8, Swin-Transformer, and ConvNeXt, from the experiment, it can be seen that our YOLO-C3AC algorithm has an average accuracy of 80.2%, which is higher than the above detectors. This indicates that our model can achieve efficient detection of kidney stones.

\* Corresponding author.



This work is licensed under a Creative Commons Attribution International 4.0 License.

BDCTA 2025, Kuala Lumpur, Malaysia  
© 2025 Copyright held by the owner/author(s).  
ACM ISBN 979-8-4007-1362-0/2025/02  
<https://doi.org/10.1145/3727505.3727510>

## CCS Concepts

• **Computing methodologies** → Artificial intelligence; Computer vision..

## Keywords

Deep Learning, Object Detection, CT Image, YOLO, Kidney Stone

### ACM Reference Format:

Jingqing Tang, Yonghong Li, and Juan Wang. 2025. YOLO-C3AC: An Efficient Deep Learning Method for CT Image Detection of Kidney Stones. In *2025 International Conference on Big Data, Communication Technology and Computer Applications (BDCTA 2025)*, February 14–16, 2025, Kuala Lumpur, Malaysia. ACM, New York, NY, USA, 8 pages. <https://doi.org/10.1145/3727505.3727510>

## 1 Introduction

Nephrolithiasis is a condition in which pebbles-like structures form in the kidneys due to the accumulation of minerals. These include calcium oxalate, guanec, uric acid, and cystine stones. Stone formation involves nucleation, growth, aggregation, and retention within the kidneys, which are influenced by both internal factors (age, sex, genetics) and external factors (geography, diet) [1]. The incidence of kidney stones is increasing globally [2] [3]. In China, the prevalence is 5.8% (6.5% in males and 5.1% in females) [4]. In the United States, the incidence is on the rise, especially in men [5]. Cardiovascular disease, chronic kidney disease, diabetes, etc. will affect its occurrence [6]. Symptoms include hematuria, renal colic, dull aches, nausea, and urgency [7]. The technology for stone detection includes imaging techniques such as ultrasound, MRI, CT scanning, etc. CT scans are commonly used because they are informative [8]. However, doctors can also make mistakes when they manually test for kidney stones. AI technology can solve this problem [9]. Modern deep learning algorithms, such as the YOLO series, excel in this task [10] [11]. In the detection of kidney stones, techniques such as median filtering, morphological manipulation, and neural networks have been used [12] [13]. However, these traditional testing methods often only detect the presence of stones, not specific locations or sizes. YOLOv7 and YOLOv5 have greater potential for medical imaging detection, providing more precise results [14].

In the CT dataset of kidney stone detection, the image resolution is poor, the organs or tissues are overlapped, and the target stones are small and distributed in a clustered or discrete manner. Our YOLO-C3AC based on YOLOv5 aims to address these issues and improve detection performance, and the following improvements

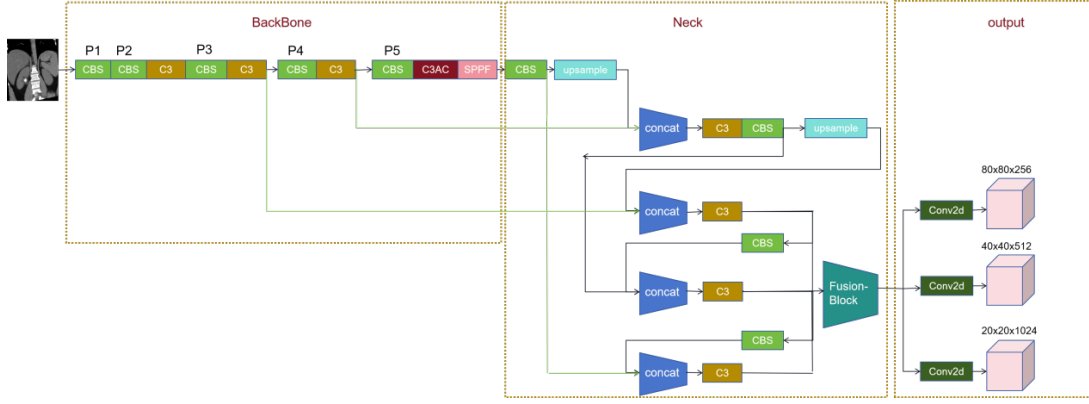


Figure 1: The structure of YOLO-C3AC

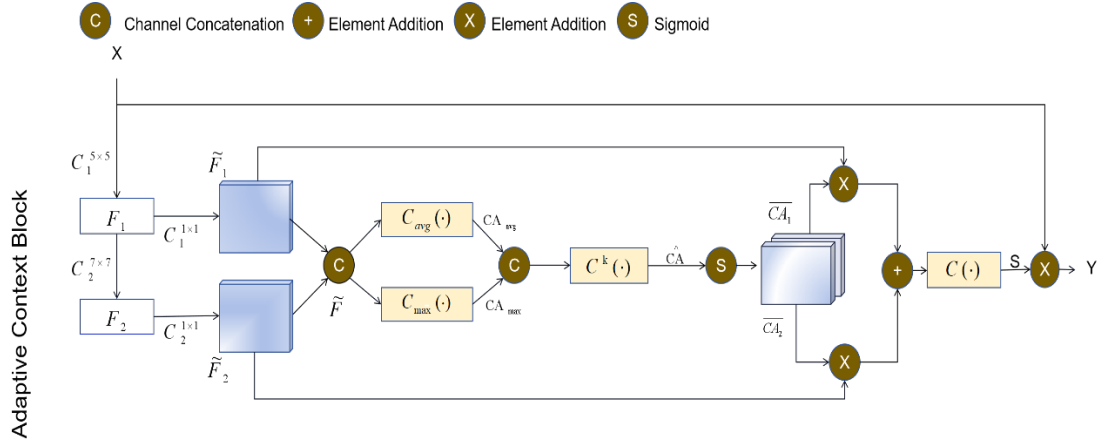


Figure 2: ACBlock structure

were emphatically improved. 1. Replacing the last C3 module of YOLOv5 backbone network with C3AC module can enable the model to obtain adaptive size information, thereby improving the ability to accurately locate kidney stones. 2. Add FusionBlock in front of the detection head to improve the spatial position information perception performance of the model. 3. The MPDIoU loss function is used, which can speed up the model convergence and improve its detection accuracy. 4. Soft NMS can reduce the probability of missed and false detections and improve detection accuracy.

## 2 Design of YOLO-C3AC network

YOLO-C3AC network architecture represents an enhancement based on the YOLOv5 framework, considering the computational complexity and the fact that high-level features have more semantic features. We only modify the bottleneck module in the last C3 of the backbone network to the C3AC module, adding adaptive contextual information for advanced features. FusionBlock module was added before the detection head, enhancing the model's ability to perceive spatial positions at different scales. We use the latest MPDIoU loss function in the model to accelerate model convergence.

NMS(Non-Maximum-Suppression) is used to remove duplicate detection results and only retain the best target box, and thereby improves model accuracy and efficiency. Soft-NMS gradually reduces the scores of overlapping boxes by modifying the scores. we choose to integrate Soft-NMS in the network in order to simplify the post-processing process and avoid introducing additional parameters. Figure 2 shows the model structure of YOLO-C3AC.

Model structure constructed in this article is shown in Figure 1.

### 2.1 Backbone network structure design

According to CondConv [15], Dynamic Convolution [16] and LSKNet [17], we propose the ACBlock module, which dynamically adjusts the model's receptive field for better contextual information processing. By replacing the last bottleneck of C3 in YOLOv5 backbone network with ACBlock (see Figure 2), now termed the C3AC module see Figure 3 (CBS means Conv + BN + SiLU), the model gains enhanced contextual information for detecting small samples in complex backgrounds.

The ACBlock module dynamically determines convolutional kernel weights based on input, effectively utilizing two large kernels. It adjusts target receptive fields spatially, merging them for optimal

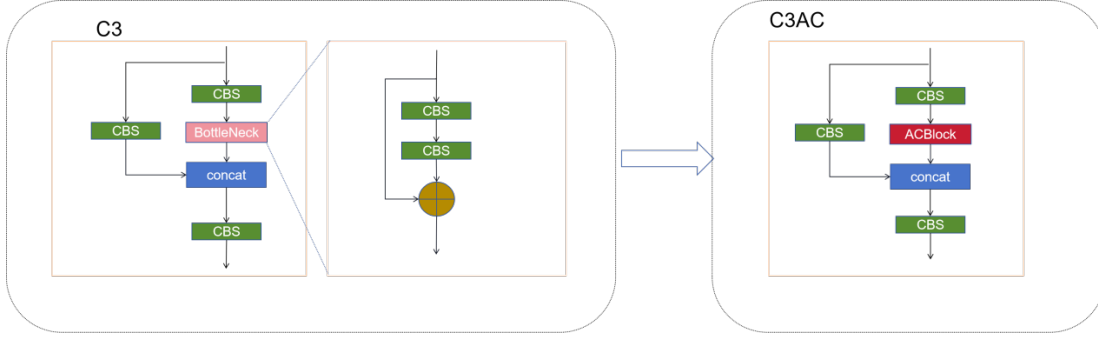


Figure 3: C3 module and C3AC module structure

space utilization. To integrate adaptive contextual information into the kidney stone detection model without parameter increase, the paper employs sequential decomposition convolution with varied receptive fields, illustrated in Formula (1).

$$F_0 = X, \quad F_{i+1} = C_i^k(F_i) \quad (1)$$

Where  $C_i^k(\cdot)$  is a two-dimensional convolution with kernel size  $k$ ,  $i$  represents the sequence number of the current convolution operation.

The input of kidney stone detection undergoes convolution (5x5) to yield  $F_1$ , and  $F_1$  is further subjected to convolution (7x7) to generate  $F_2$ . Then 1x1 convolution is applied to these two results to obtain feature vectors  $\tilde{F}_1$  and  $\tilde{F}_2$ , respectively. It formed two branches. One branch combines two feature vectors through channel blending and yields  $\tilde{F} = [\tilde{F}_1; \tilde{F}_2]$ . The other branch serves as the corresponding large kernel sequence feature to prepare for the subsequent spatial selection mask weighting. Then, a spatial dynamic selection kernel operation will be performed by  $\tilde{F}$ , which can improve the network's focus on crucial spatial context regions for precise target detection. This operation is implemented as follows. Firstly, the spatial relationship will be effectively extracted by maximizing the average value based on the channel (represented as  $C_{avg}(\cdot)$  and  $C_{max}(\cdot)$ ):

$$CA_{avg} = C_{avg}(\tilde{F}), \quad CA_{max} = C_{max}(\tilde{F}) \quad (2)$$

Where  $CA_{avg}$  and  $CA_{max}$  are the average and maximum operators.

$$\hat{CA} = C^k([CA_{avg}; CA_{max}]) \quad (3)$$

Then, these two features are connected in the spatial dimension and a spatial attention map  $\hat{CA}$  by a convolution  $C^k(\cdot)$  is obtained, which allows for information exchange between different spaces. Then, sigmoid function can activate each channel dimension of the spatial attention map  $\hat{CA}_i$  to obtain its corresponding spatial selection mask:

$$\overline{CA}_i = \sigma\left(\hat{CA}_i\right) \quad (4)$$

Where  $\sigma(\cdot)$  is the sigmoid function. Then, the features of the large kernel sequence retained by the branch are weighted according to their corresponding spatial selection masks, and are fused through

convolutional layers  $C(\cdot)$  to obtain attention features  $S$ :

$$S = C\left(\sum_{i=1}^N (\overline{CA}_i \cdot \tilde{F}_i)\right) \quad (5)$$

The final output of ACBlock  $Y$  is the product of the elements between the input features  $X$  and  $S$ :

$$Y = X \cdot S \quad (6)$$

In the C3AC module, the ACBlock module provides adaptive contextual information for model detection, which helps the model accurately locate kidney stones in the kidney, thereby improves the accuracy of model detection.

## 2.2 FusionBlock construct

Inspired by the Dynamic Head [18], PANNet [19], Deformable ConvNets [20], Mask R-CNN [21], FOCS [22] and RepPoints [23], we developed the FusionBlock module to enhance the kidney stone detection model's performance. Figure 4 shows the FusionBlock structure.

Feature maps from input modules of three scales are uniformly represented as  $F_i \in R^{H \times W \times C}$ , where  $W$ ,  $H$  represent the width, height, and  $C$  represent Number of channels. Due to the various geometric transformations and spatial features of the stone samples in CT images of kidney stones, the improvement of the information obtained from different spatial positions of  $F_i$  can enhance the model's spatial perception when detecting kidney stone samples. Firstly, the feature maps from three scales are convolved 1x1 and connected in the channel dimension to obtain  $\tilde{F}$ . The spatial attention calculation (following deformable convolution operations) is performed on  $\tilde{F}$  to obtain  $\tilde{F}$ :

$$\tilde{F} = \pi(\cdot) = \sum_{k=1}^K w_k \cdot \tilde{F}(p_k + \Delta p_k; c) \cdot \Delta m_k \quad (7)$$

$K$  represent the total number of locations where sparse sampling is performed,  $p_k + \Delta p_k$  represent the shift position of self-learning spatial offset,  $\Delta p_k$  represents the offset ratio,  $\Delta m_k$  is the scalar of self-learning discrimination at the position  $p_k$ ,  $\Delta p_k$  and  $\Delta m_k$  both are learned from the features of  $\tilde{F}$ .

After performing slice operations on the obtained data  $\tilde{F}$ , feature maps  $F_i$  corresponding to three scales are obtained and enter the detection head. the detection model can improve its ability to perceive spatial positions at different scales. Due to the significant randomness in the shape and position of stone samples in the image,

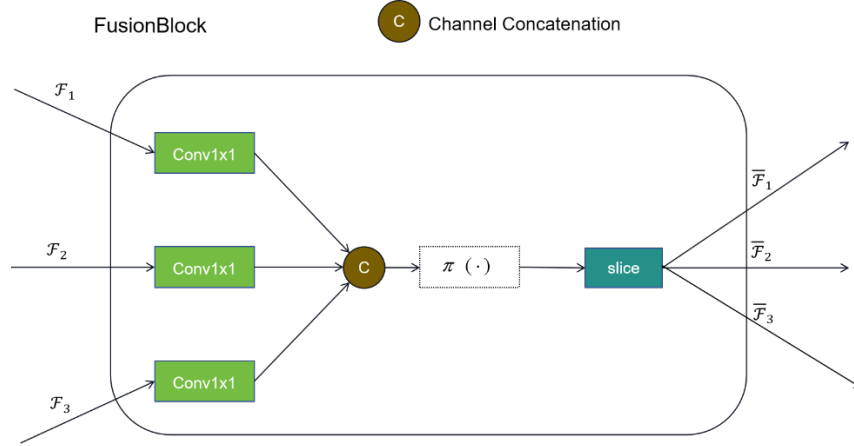


Figure 4: FusionBlock module

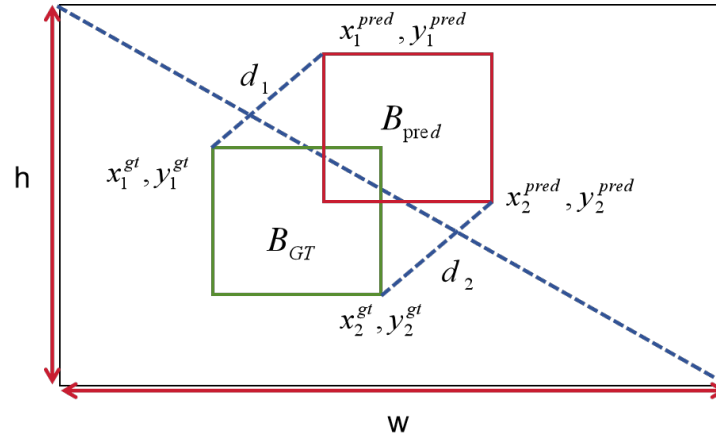


Figure 5: MPDIoU calculation

adding FusionBlock can improve the model's ability to accurately detect stone samples.

### 2.3 Integrated MPDIoU loss function

Currently, object detectors use bounding box regression (BBR) for target localization. Two-stage detectors like CascadeRcnn [24] and one-stage detectors like the YOLO series (YOLOv3, YOLOv7, YOLOv8) and SSD series (DSSD [25]) employ BBR to improve accuracy and inference speed, respectively. IoU-based loss functions are used to evaluate localization accuracy by calculating the overlap between predicted and actual bounding boxes. Figure 5 visualizes the position of the variable when calculating the MPDIoU.

In the computation process, the initial step involves processing the current label data to extract the coordinates  $(x_1^{pred}, y_1^{pred})$ ,  $(x_2^{pred}, y_2^{pred})$  of top-left and bottom-right corners for the predicted bounding box  $B_{pred}$ . Simultaneously, coordinates  $(x_1^{gt}, y_1^{gt})$ ,  $(x_2^{gt}, y_2^{gt})$  of corners for the ground truth box  $B_{gt}$  are obtained. Subsequently, the square value of the distance between the

top-left and bottom-right corners is calculated:

$$d_1^2 = (x_1^{pred} - x_1^{gt})^2 + (y_1^{pred} - y_1^{gt})^2 \quad (8)$$

$$d_2^2 = (x_2^{pred} - x_2^{gt})^2 + (y_2^{pred} - y_2^{gt})^2 \quad (9)$$

$$MPDIou = \frac{|B_{gt} \cap B_{pred}|}{|B_{gt} \cup B_{pred}|} - \frac{d_1^2}{w^2 + h^2} - \frac{d_2^2}{w^2 + h^2} \quad (10)$$

$$L_{MPDIoU} = 1 - MPDIou \quad (11)$$

The MPDIoU loss function enhances boundary regression accuracy, leading to improved model detection precision and accelerated convergence speed.

### 2.4 Add Soft-NMS to the model

Traditional NMS adopts a re-evaluation function when dealing with overlapping box scores:

$$s_i = \begin{cases} s_i, & iou(M, b_i) < N_t, \\ 0, & iou(M, b_i) \geq N_t. \end{cases} \quad (12)$$

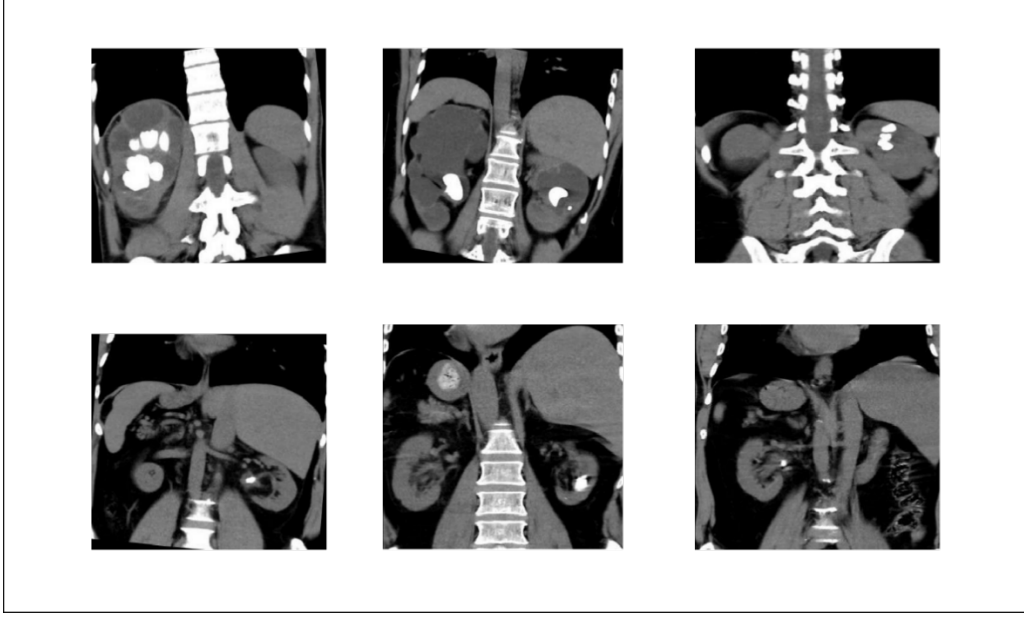


Figure 6: Kidney stone sample

$N_t$  represents the NMS threshold,  $M$  represents the detection box that achieves the highest confidence score.

Soft NMS is achieved through a modified re-scoring function:

$$s_i = \begin{cases} s_i, & iou(M, b_i) < N_t, \\ s_i(1 - iou(M, b_i)), & iou(M, b_i) \geq N_t. \end{cases} \quad (13)$$

Redefined function transforms detection scores above the threshold into a linear function overlapping with  $M$  values, penalizing closely positioned boxes more. Soft NMS can reduce the possibility of errors and missed detections, thereby improving the accuracy and efficiency of the detection process.

### 3 Experiment result analysis

#### 3.1 Experiment data

We adopts a medical image set from the Kaggle platform which is specifically designed to detect kidney stones using advanced object detection methods. This dataset contains various kidney stone images from coronary CT scans, which covers different sizes, shapes, and locations of stones in the urinary system. This dataset contains 1300 CT images of kidney stones with approximately 3000 kidney stone samples and corresponding label information. Each image has clear bounding boxes with a pixel size of  $391 \times 320$ . This dataset contains 1054 images in the training set with 2278 samples, There are 325 samples and 123 images in the validation set and 224 samples and 123 images in the test. kidney stone samples are shown in Figure 6 (with white highlights indicating stones).

#### 3.2 Experimental evaluation indexes

We use mAP50 to evaluate model performance. mAP50 refers to when the IoU threshold is 0.5. It is an average accuracy. mAP0.5:0.95 is IoU threshold increases from 0.5 to 0.95 with an increase step size of 0.05. When calculating its average value at different IoU

thresholds, it is called mAP. The experiment mainly used mAP50 and mAP0.5:0.95 to evaluate model.

#### 3.3 Experiment results

Figure 7 shows comparison of thermal maps between YOLO-C3AC and YOLOv5 in detecting CT images of kidney stones in the test set. In Figure 8, due to the implementation of early stopping based on convergence criteria in the experiments, the YOLO-C3AC ceased training prematurely after 230 epochs. Figure 9 shows comparison of the detection results. we found that the original YOLOv5s model had a false detection problem when identifying kidney stone samples, mistakenly identifying bright spots on other organs as stones. In contrast, our proposed model successfully avoids this problem and has higher confidence on the same detection target. This indicates that our improvement plan is effective.

Table 1 shows the final comparison results of the mAP50 indicators of several models

#### 3.4 Ablation experiment

Table 2 shows the experimental results.

From the above results, we can conclude that:

After adding C3AC module, the mAP50 Improved 1.9%, and the mAP0.5:0.95 Improved 0.2%, indicating that the module we proposed to extract adaptive size context information is effective.

After integrating the FusionBlock module, the mAP50 Improved 2.4%, and the mAP0.5:0.95 Improved 2.8%, indicating that the FusionBlock with spatial awareness can be fused with feature maps of different scales, which improves the performance of the model.

After introducing the MPDIoU function, the mAP50 Improved 0.9%, and the mAP0.5:0.95 Improved 0.8%. indicating the new function can enhance detection performance, improve the detection accuracy of model.

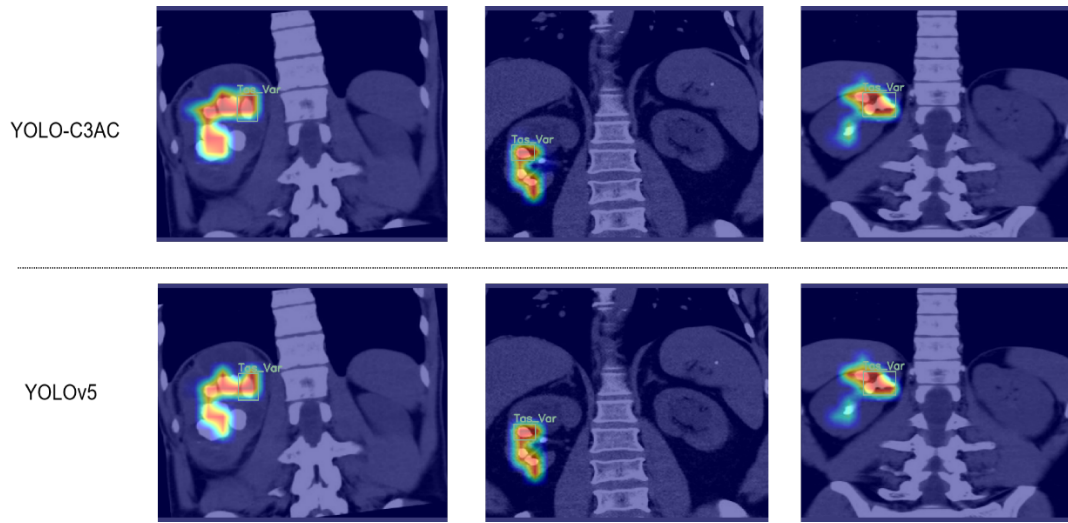


Figure 7: thermal maps compare

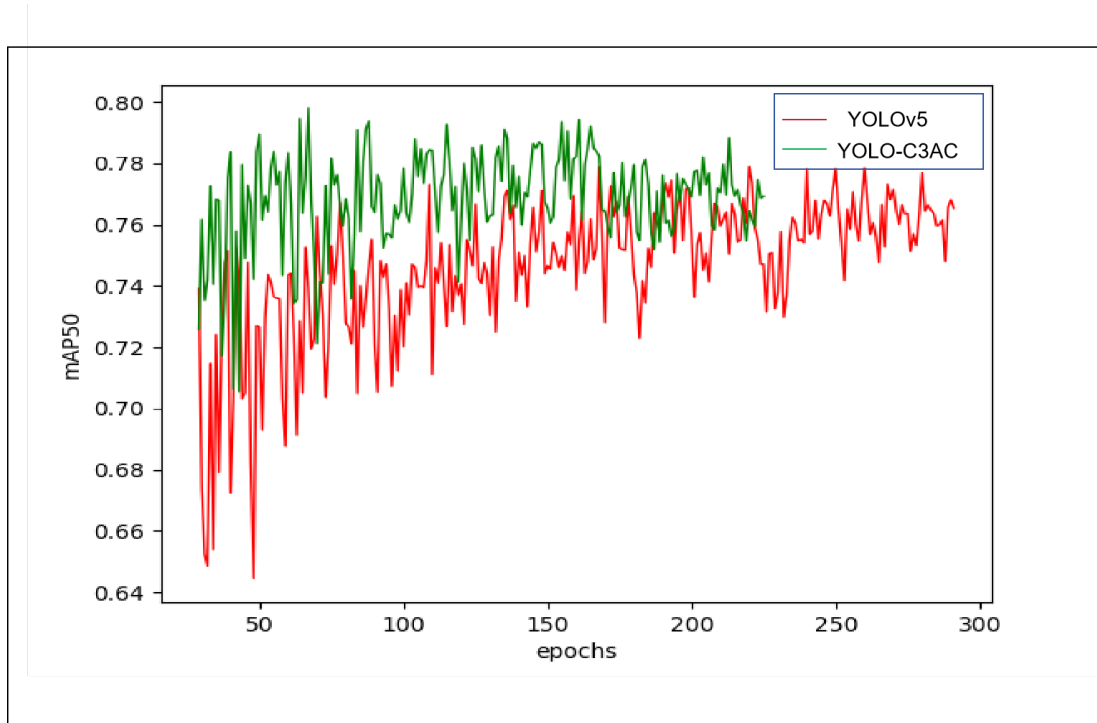


Figure 8: mAP50 compare

Introducing Soft NMS increased mAP50 by 2.5% and mAP 0.5:0.95 by 3.2%.

The ablation experiment not only verified that the proposed improvements had improved the kidney stone detection model, but also that it was wise to combine these improvements.

#### 4 Conclusion

The YOLO-C3AC model solves the limitations of traditional machine learning and general deep learning techniques that are difficult to handle complex scenarios to improve kidney stone detection. The proposed model combines the C3AC module for adaptive contextual information, FusionBlock for integrating multi-scale features,



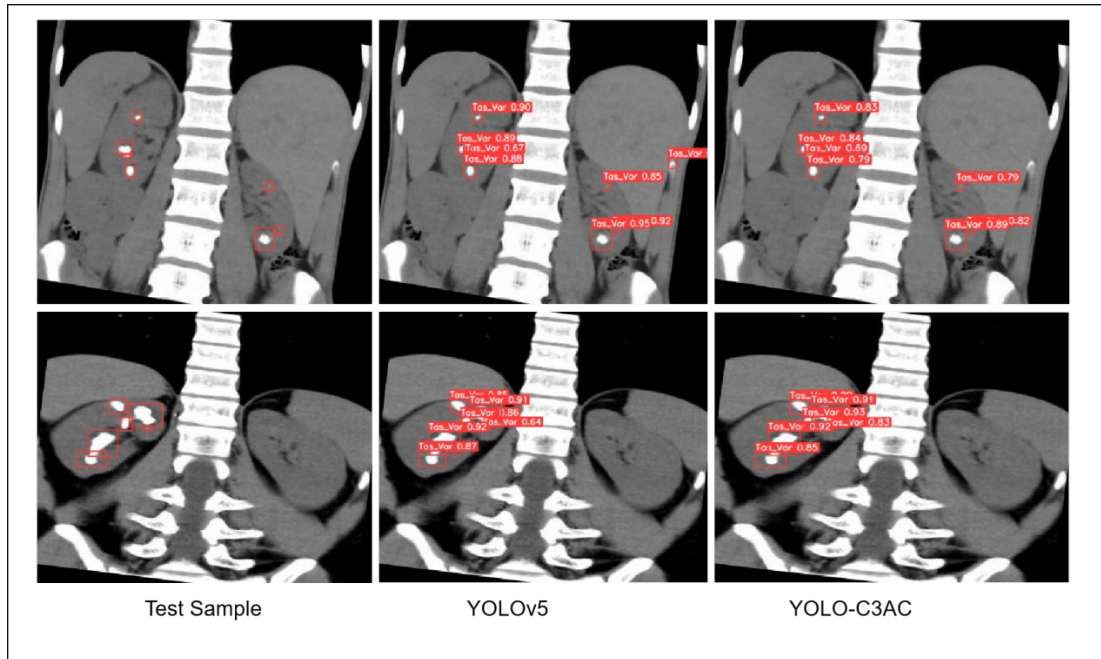


Figure 9: .Comparison of YOLO-C3AC vs YOLOv5 detection results

Table 1: Contrast experiment.

| Model      | mAP50 |
|------------|-------|
| YOLOv7     | 0.747 |
| YOLOv5     | 0.751 |
| YOLOv3     | 0.743 |
| YOLOv8     | 0.778 |
| YOLO-C3AC  | 0.802 |
| Swin-T     | 0.754 |
| ConvNeXt-T | 0.762 |

Table 2: Contrast test

| Model              | C3AC | FusionBlock | MPDIoU | Soft-NMS | mAP50 | mAP0.5:0.95 |
|--------------------|------|-------------|--------|----------|-------|-------------|
| YOLOv5             | ×    | ×           | ×      | ×        | 0.751 | 0.314       |
| YOLOv5+C3AC        | ✓    | ×           | ×      | ×        | 0.770 | 0.316       |
| YOLOv5+FusionBlock | ×    | ✓           | ×      | ×        | 0.775 | 0.342       |
| YOLOv5+MPDIoU      | ×    | ×           | ✓      | ×        | 0.760 | 0.322       |
| YOLOv5+Soft-NMS    | ×    | ×           | ×      | ✓        | 0.776 | 0.346       |
| YOLO-C3AC          | ✓    | ✓           | ✓      | ✓        | 0.802 | 0.378       |

which improves spatial position perception and detection accuracy of the model. Use MPDIoU to optimize model closure and Soft NMS to reduce errors and missed detections. However, the model has limitations, such as not using the Transformer method for small object detection and not using a center point to represent the target. The future will include the use of higher-quality CT data,

data augmentation, and the exploration of new ways to improve performance.

## 5 Code and Data Availability:

The data supporting the results of this study can be publicly obtained on the following URL/DOI: <https://www.kaggle.com/datasets/safurahajiheidari/kidney-stone-images/code>.

## Acknowledgments

Our work has received support from the National Natural Science Foundation of China (Grant No.2020YFC2003502 61876201,62136002,61901074, 61972060, 72171031), China Postdoctoral Science, Foundation (Grant No. 2021M693771) and the Natural Science Foundation Project of Chongqing (Grant No. cstc2020jcyjmsxmX0649) and Nan'an District Joint Science and Health Medical Research Project of Chongqing (2024-21).

## References

- [1] Rahimnia R, Mohammadi A, Pakdel A, *et al.* Molecular biology, genetic, and epigenetic urolithiasis[M]//Genetics and Epigenetics of Genitourinary Diseases. Academic Press, 2025: 139-149.
- [2] Chen K, Meskawi M, Miller L E, *et al.* Trends in kidney stone prevalence among US adults: A concerning contemporary gender analysis from the NHANES database[J]. Canadian Urological Association Journal, 2025, 19(2).
- [3] Qin K, Qing J, Wang Q, *et al.* Epidemiological shifts in chronic kidney disease: a 30-year global and regional assessment[J]. BMC Public Health, 2024, 24(1): 3519.
- [4] Zeng, G., Mai, Z., Xia, S., Wang, Z., Zhang, K., Wang, L., . . . & Ye, Z. (2017). Prevalence of kidney stones in china: an ultrasonography based cross-sectional study. BJU International, 120(1), 109-116. <https://doi.org/10.1111/bju.13828>.
- [5] Chewcharat, A., Curhan, G. Trends in the prevalence of kidney stones in the United States from 2007 to 2016. Urolithiasis 49, 27–39 (2021). <https://doi.org/10.1007/s00240-020-01210-w>.
- [6] Stamatelou, K.; Goldfarb, D.S. Epidemiology of Kidney Stones. *Healthcare* **2023**, *11*, 424. <https://doi.org/10.3390/healthcare11030424>.
- [7] Curhan G C, Aronson M D, Preminger G M. Kidney stones in adults: Diagnosis and acute management of suspected nephrolithiasis[J]. UpToDate. Waltham, MA: UpToDate, 2023.
- [8] Asif S, Zheng X, Zhu Y. An optimized fusion of deep learning models for kidney stone detection from CT images[J]. Journal of King Saud University-Computer and Information Sciences, 2024, 36(7): 102130.
- [9] Yenikekaluva A, Azeez S F, Sakegaonkar A, *et al.* UrologiQ: AI-based accurate detection, measurement and reporting of stones in CT-KUB scans[J]. Urolithiasis, 2024, 52(1): 170.
- [10] Xu T, Jiang J, Cai L, *et al.* Salient object detection with non-local feature enhancement and edge reconstruction[J]. Scientific Reports, 2025, 15(1): 397.
- [11] Abdigapporov S, Miraliev S, Kakani V, *et al.* Joint multiclass object detection and semantic segmentation for autonomous driving[J]. IEEE Access, 2023, 11: 37637-37649.
- [12] Shangguan Z, Rostami M. Improved region proposal network for enhanced few-shot object detection[J]. Neural Networks, 2024, 180: 106699.
- [13] Wang A, Chen H, Liu L, *et al.* Yolov10: Real-time end-to-end object detection[J]. arXiv preprint arXiv:2405.14458, 2024.
- [14] Zhang H, Liu K, Gan Z, *et al.* UAV-DETR: Efficient End-to-End Object Detection for Unmanned Aerial Vehicle Imagery[J]. arXiv preprint arXiv:2501.01855, 2025.
- [15] Yang B, Bender G, Le Q V, *et al.* Condconv: Conditionally parameterized convolutions for efficient inference[J]. Advances in neural information processing systems, 2019, 32.
- [16] Chen Y, Dai X, Liu M, *et al.* Dynamic convolution: Attention over convolution kernels[C]//Proceedings of the IEEE/CVF conference on computer vision and pattern recognition. 2020: 11030-11039.
- [17] Li Y, Hou Q, Zheng Z, *et al.* Large Selective Kernel Network for Remote Sensing Object Detection[J]. arXiv preprint arXiv:2303.09030, 2023.
- [18] DAI X, CHEN Y, XIAO B, *et al.* Dynamic Head: Unifying Object Detection Heads with Attentions[C/OL]//2021 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), Nashville, TN, USA. 2021. <http://dx.doi.org/10.1109/cvpr46437.2021.00729>. DOI:10.1109/cvpr46437.2021.00729.
- [19] Ruppel F, Faion F, Gläser C, *et al.* Transformers for object detection in large point clouds[C]//2022 IEEE 25th International Conference on Intelligent Transportation Systems (ITSC). IEEE, 2022: 832-838.
- [20] Weiguang L I U, Dong L I U, Lu W. Survey of Deformable Convolutional Networks[J]. Journal of Frontiers of Computer Science & Technology, 2023, 17(7).
- [21] Rubin R, Jacob C, Anzar S M, *et al.* Mask R-CNN with Multi-Backbones-A Comparative Analysis[C]//2022 5th International Conference on Signal Processing and Information Security (ICSPIS). IEEE, 2022: 55-60.
- [22] Chu Y, Guo J, Shan W, *et al.* EfficientFCOS: An efficient one-stage object detection model based on FCOS[C]//2022 IEEE 25th International Conference on Computer Supported Cooperative Work in Design (CSCWD). IEEE, 2022: 617-622.
- [23] Song J, Miao L, Zhou Z, *et al.* Optimized point set representation for oriented object detection in remote sensing images[J]. IEEE Geoscience and Remote Sensing Letters, 2023.
- [24] Huang H, Li L, Ma H. An improved cascade R-CNN-Based target detection algorithm for UAV Aerial Images[C]//2022 7th International Conference on Image, Vision and Computing (ICIVC). IEEE, 2022: 232-237.
- [25] Wu T, Sha T, Yao X, *et al.* Improvement of the YOLO Series for Detecting Tower Cranes Based on High-Resolution Remote Sensing Imagery[J]. Journal of Geovisualization and Spatial Analysis, 2025, 9(1): 8.